CYBERSECURITY LOG ANALYSIS REPORT

Student Answer Template

Name: Mark Daniel M. Patricio
Activity: CTM-ACTIVITY-8
Date: October 31, 2026

ATTACK 1 ANALYSIS

1. **Identification of Attack** Brute Force Attack (SSH Dictionary Attack)
2. **Evidence from the System Logs**

Evidence 1: Event ID / Timestamp: BF-0001 / 2026-10-31T08:14:02Z

Observed activity: SSH authentication failure for the username 'admin' from source IP 198.51.100.23.

Why it matters: This indicates the initial unauthorized attempt to gain remote access to the system using a common default administrator username.

Evidence 2: Event ID / Timestamp: BF-0003 / 2026-10-31T08:14:12Z
Observed activity: A failed login attempt using a different username ('administrator') from the exact same source IP address.
Why it matters: It demonstrates that the attacker is systematically cycling through a list of standard usernames, which is a clear indicator of an automated brute-force or dictionary attack.

Evidence 3: Event ID / Timestamp: BF-0005 / 2026-10-31T08:14:22Z
Observed activity: The 'admin' account was temporarily locked, and the status changed to 'blocked'.
Why it matters: This shows that the system's security policies correctly identified the rapid sequence of failed logins (5 attempts in just 20 seconds) and engaged automated defensive measures.

3. Attack Analysis

A threat actor using the IP address 198.51.100.23 targeted the system at 192.0.2.10 over port 22 (SSH). The attacker rapidly attempted to authenticate using standard administrative usernames (admin, administrator, root) in a tight span of 20 seconds. This automated cycling of credentials is a classic brute-force attack aimed at guessing passwords to gain remote command-line access.

4. Attack Status

Unsuccessful. The system's security controls functioned as intended. The repeated failed logins triggered an account lockout (BF-0005) and subsequently rate-limited the attacker's IP address (BF-0007). While a successful login occurred later (BF-0008), it originated from a

completely different IP (203.0.113.44) using a standard 'student' account, indicating a separate, legitimate user login.

5. Possible Scenario if the Attack Succeeds

If the attacker had successfully guessed the password for 'root' or 'admin', they would have gained complete remote administrative control over the targeted server. This would allow them to steal sensitive data, install malware or backdoors, modify system configurations, and use the compromised machine to pivot and attack other systems within the internal network.

6. Prevention and Mitigation

Recommendation 1: Disable direct root login over SSH in the system configuration file.
Recommendation 2: Implement Multi-Factor Authentication (MFA) for all remote SSH access.
Recommendation 3: Configure intrusion prevention tools to automatically block IP addresses that generate excessive authentication failures.

ATTACK 2 ANALYSIS

1. Identification of Attack

Phishing Attack (Credential Harvesting and Malicious Attachment)

2. Evidence from the System Logs Evidence 1: Event ID / Timestamp: PH-0001 / 2026-10-31T09:02:11Z
Observed activity: An email titled "Urgent: Password Expiration" was received, containing a URL domain (account-verify.example.test) that did not match the official organization domain.
Why it matters: This displays the initial delivery of a social engineering lure designed to create urgency and trick the user into clicking a deceptive link.

Evidence 2: Event ID / Timestamp: PH-0003 / 2026-10-31T09:05:18Z
Observed activity: The targeted user (employee01) clicked the link that was previously flagged as suspicious.
Why it matters: This indicates that the psychological manipulation of the phishing email worked, moving the attack into the active execution phase by the user.

Evidence 3: Event ID / Timestamp: PH-0004 / 2026-10-31T09:05:41Z
Observed activity: The endpoint browser security protection blocked access to a page resembling a credential form.
Why it matters: It confirms the attacker's goal was to harvest the user's login credentials, but technical security controls successfully intervened to prevent data loss.

3. Attack Analysis

An attacker operating from IP 203.0.113.80 spoofed an internal IT support address to send a deceptive email regarding "Password Expiration." The goal was to trick employee01 into entering their credentials on a fake login page. A second wave of the attack targeted employee02 using a spoofed payroll email containing a malicious HTML attachment. This indicates a targeted phishing campaign using multiple lures (urgency and financial curiosity).

4. Attack Status

Unsuccessful. Although the attack successfully bypassed initial delivery filters and employee01 clicked the malicious link (PH-0003), the overall attack failed. Endpoint browser protection blocked the credential harvesting form (PH-0004), and the email security gateway correctly blocked the malicious HTML attachment sent to employee02 (PH-0007). No systems or credentials were compromised.

5. Possible Scenario if the Attack Succeeds

If employee01 had submitted their credentials, the attacker could have taken over their account. This would grant the attacker access to corporate emails, internal documents, and potentially the ability to move laterally across the network or send further internal phishing emails that appear highly credible to other staff.

6. Prevention and Mitigation

Recommendation 1: Implement mandatory, regular cybersecurity awareness training focusing on identifying spoofed domains and verifying urgent IT requests.
Recommendation 2: Enforce email authentication protocols such as SPF, DKIM, and DMARC to prevent attackers from successfully spoofing internal company addresses.
Recommendation 3: Maintain strict email gateway policies that automatically quarantine executable or suspicious HTML attachments.

ATTACK 3 ANALYSIS

1. Identification of Attack

Distributed Denial of Service (DDoS) Attack

2. Evidence from the System Logs

Evidence 1: Event ID / Timestamp: DOS-0002 / 2026-10-31T10:20:10Z
Observed activity: The request rate from IP 198.51.100.90 spiked to 420 requests per second, far exceeding the baseline of 35, which triggered rate limiting.
Why it matters: This abnormal surge in traffic demonstrates a deliberate attempt to overwhelm the web server's processing capabilities.

Evidence 2: Event ID / Timestamp: DOS-0003 / 2026-10-31T10:20:20Z
Observed activity: Elevated traffic spikes were recorded from additional source IPs (e.g., 198.51.100.91 and .92).
Why it matters: The involvement of multiple, distinct IP addresses confirms the "distributed" nature of the attack, making it harder to block a single source.

Evidence 3: Event ID / Timestamp: DOS-0004 / 2026-10-31T10:20:30Z
Observed activity: Service latency heavily increased, and the server began returning HTTP 503 (Service Unavailable) response codes.
Why it matters: This proves the attack temporarily succeeded in exhausting the server's resources, directly denying access to legitimate traffic.

3. Attack Analysis

A coordinated attack utilizing multiple IP addresses flooded the target web server (192.0.2.20) on port 443. The traffic volume rapidly escalated from a normal baseline of 35 requests per second up to a massive 850 requests per second. The sheer volume of incoming HTTPS requests overwhelmed the application, causing latency and health check failures before automated mitigation finally kicked in.

4. Attack Status

Partially Successful. The attack managed to temporarily disrupt the availability of the web server, forcing it to drop requests and return 503 Service Unavailable errors (DOS-0004). However, the attack did not result in a total, prolonged outage because automated traffic filtering and rate mitigation successfully engaged (DOS-0006), dropping the traffic back to manageable levels and restoring service.

5. Possible Scenario if the Attack Succeeds

If the mitigation had failed and the attack succeeded long-term, the organization's web applications would experience prolonged downtime. This would result in a complete disruption of digital services, loss of revenue, damaged customer trust, and severe operational delays until the traffic subsided or was manually blocked.

6. Prevention and Mitigation

Recommendation 1: Deploy a specialized Web Application Firewall (WAF) and a cloud-based DDoS protection service to scrub malicious traffic.
Recommendation 2: Configure load balancers and auto-scaling groups to automatically provision more resources to absorb unexpected traffic spikes.
Recommendation 3: Implement aggressive geographic blocking and behavioral rate-limiting to drop traffic that doesn't match standard user patterns.

FINAL REFLECTION

1. Among the three attacks, which could cause the greatest damage to an organization? Explain your answer.

A successful Phishing attack (Attack 2) could cause the greatest long-term damage. While a DDoS attack disrupts availability, a compromised employee account from a phishing attack gives the threat actor a foothold inside the internal network. From there, they can bypass perimeter defenses, escalate privileges, deploy destructive ransomware, or exfiltrate highly confidential company data—all while masquerading as a trusted user.

2. Why is analyzing system logs important in detecting and responding to cybersecurity incidents?

Analyzing system logs provides the definitive "ground truth" of a network's activity. Without logs, security teams are operating blindly. Logs allow analysts to determine exactly how an attacker entered, what systems they touched, whether data was compromised, and how to properly patch the vulnerability to prevent a recurrence rather than just guessing.

3. What is the most important cybersecurity lesson you learned from this activity?

The most important lesson is the necessity of "Defense in Depth" (layered security). In the phishing scenario, a human error occurred when the user clicked the malicious link. However, because endpoint browser protection and email gateways were also in place, the attack was stopped before a compromise occurred. Relying on a single line of defense is never enough.